CHAPTER 6

Medium Access Control

ALOHA · slotted ALOHA · CSMA · CSMA/CD · binary exponential backoff · minimum frame size

QUICK TRICK

Pure ALOHA: $\mathbf{S = G\,e^{-2G}}$, max **18.4%** at G = 0.5. Slotted ALOHA: $\mathbf{S = G\,e^{-G}}$, max **36.8%** at G = 1. CSMA/CD needs $\mathbf{T_t \geq 2T_p}$, i.e. min frame $\mathbf{L = 2 \times T_p \times B}$. Backoff: after the i-th collision pick K from $\mathbf{0 \ldots 2^i - 1}$.

6.1 The problem

● **LEVEL 1 · BEGINNER**

Many stations share one channel (a cable or radio). If two talk together, the signals collide and both are lost. A **MAC protocol** decides who talks when.

Analogy. A meeting room: everyone can speak, but if two speak together nothing is understood. Rules: speak at fixed times (slotted), listen first (CSMA), stop if you hear someone else (CD).

Family	Examples	Idea
Random access	ALOHA, CSMA, CSMA/CD	Send when you like; handle collisions
Controlled access	Polling, token ring	Permission to send (no collisions)
Channelisation	TDMA, FDMA, CDMA	Split the channel in time / frequency / code

6.2 ALOHA

● **LEVEL 2 · EXAM**

Let T = frame time, G = average frames offered per frame time, S = successful frames per frame time (throughput). Each frame that overlaps another is destroyed.

	Pure ALOHA	Slotted ALOHA
When to send	Any time	Only at the start of a slot (slot = T)
Vulnerable time	2T	T
Throughput S	$G\,e^{-2G}$	$G\,e^{-G}$
Maximum	1/(2e) ≈ 0.184 at G = 0.5	1/e ≈ 0.368 at G = 1

Example. Slotted ALOHA on a 10 Mbps channel at G = 1: S = 0.368, throughput = **3.68 Mbps**. At G = 2: S = $2e^{-2}$ = 0.2707.

6.3 CSMA: listen before you talk

● **LEVEL 2 · EXAM**

Type	Channel idle	Channel busy
1-persistent	Send at once (probability 1)	Keep sensing; send as soon as idle
Non-persistent	Send at once	Wait a random time, then sense again
p-persistent	Send with probability p; with 1 − p wait one slot	Keep sensing until idle

Collisions still happen because of **propagation delay**: a station may not hear a signal that is still on its way.

6.4 CSMA/CD (classic Ethernet)

● LEVEL 3 · GATE

- Sense the channel, send, and **keep listening while sending**. On collision: stop, send a **jam signal**, back off, retry.
- **Binary exponential backoff:** after the i-th collision pick K at random from {0, 1, ..., $2^i - 1$} (the exponent is capped at 10, so at most 1023) and wait **K × slot time** (slot = 512 bit-times = 51.2 µs at 10 Mbps). Give up after 16 attempts.
- **Minimum frame size.** The sender must still be transmitting when the collision news returns, i.e. after $2T_p$ (worst case). So $\mathbf{T_t \geq 2T_p}$, giving $\mathbf{L_{min} = 2 \times T_p \times B}$.
- **Maximum length:** d_{max} = L × v / (2B). Ethernet: L = 512 bits (64 B), B = 10 Mbps.

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Station A starts at t=0;  signal reaches far station B at t = Tp.
B starts just before t = Tp  ->  collision at B; B's jam reaches A at t = 2*Tp.
A must still be sending then, else it thinks the frame was fine.  =>  Tt >= 2 Tp
```

Example. 100 Mbps, min frame 512 bits: round trip ≤ $512/10^8$ = 5.12 µs → T_p = 2.56 µs → d = 2.56 µs × 2×10^8 = **512 m**. At 1 Gbps with 512 bits: only 51.2 m, which is why Gigabit Ethernet used *carrier extension* to 4096 bits (half-duplex).

Efficiency. Textbooks give approximations such as 1/(1 + 5.44a) or 1/(1 + 6.44a) with a = T_p/T_t. Use the formula given in the question. Efficiency **falls** as a rises (long cables or short frames).

6.5 Controlled access (short)

● LEVEL 1 · BEGINNER

Polling: a controller asks each station in turn. **Token passing:** only the holder of the token may send. No collisions, bounded delay, good at heavy load, but waiting for the token wastes time at light load. (Wi-Fi CSMA/CA is not in the syllabus.)

6.6 Common mistakes

COMMON MISTAKE

- Mixing the two ALOHA formulas (pure has e^{-2G}, slotted has e^{-G}).
- Using one-way T_p instead of $2T_p$ in the minimum-frame calculation.
- Using 2^i instead of $2^i - 1$ as the largest backoff multiplier.
- Forgetting that backoff slot = 512 bit-times (not one frame time).

QUICK TRICK

Memory line: "Frame must outlast the round trip."

GATE QUICK CHECK — Chapter 6

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Pure ALOHA   : S = G e^(-2G)   max 0.184   vulnerable 2T
Slotted      : S = G e^(-G)    max 0.368   vulnerable T
CSMA/CD      : L_min = 2*Tp*B  ;  d_max = L*v/(2B)  ;  slot = 512 bit times
Backoff      : K in {0..2^i - 1} after i-th collision (cap 1023, give up at 16)
```

Practice questions

● LEVEL 1 · BEGINNER

Q1. Maximum throughput of pure ALOHA and slotted ALOHA?

Q2. Vulnerable time for pure ALOHA if the frame time is 1 ms?

Q3. After the 3rd collision, from which set does a station pick K?

Q4. Why does CSMA/CD need a minimum frame size?

● LEVEL 2 · EXAM

Q5. Minimum frame size on a 10 Mbps CSMA/CD link with one-way propagation 25.6 µs.

Q6. Maximum cable length at 100 Mbps with min frame 512 bits, v = 2×10^8 m/s.

Q7. Slotted ALOHA, 10 Mbps, G = 1. Useful throughput?

Q8. Two stations collide on their first attempt. Probability they collide again on the first retry?

● LEVEL 3 · GATE

Q9. Two stations collide on their first attempt and again on the first retry. Probability that they collide a third time (second retry)?

Q10. Slotted ALOHA with G = 2. Find S and the probability that a transmitted frame collides.

Q11. What is the largest value K can take after 12 collisions?

Q12. 1 Gbps half-duplex link, min frame 512 bits, v = 2×10^8 m/s. Maximum span?

Answers

Q	Solution
1	**18.4%** and **36.8%**.
2	2T = **2 ms**.
3	{0, 1, …, $2^3 - 1$} = **{0, …, 7}**.
4	So the sender is **still transmitting** when a collision signal returns (after $2T_p$).
5	2 × 25.6 µs × 10^7 = **512 bits** (64 B).
6	**512 m** (see 6.4).
7	0.368 × 10 = **3.68 Mbps**.
8	Each picks 0 or 1: same slot with probability **1/2**.
9	After the 2nd collision each picks K from {0, 1, 2, 3}: same value with probability **1/4**. (Probability of colliding on *both* retries from the start = 1/2 × 1/4 = 1/8.)
10	S = $2e^{-2}$ = **0.2707**. Collision probability = $1 - e^{-G} = 1 - e^{-2}$ = **0.865**.
11	The exponent is capped at 10, so K ≤ $2^{10} - 1$ = **1023**.
12	Round trip ≤ 0.512 µs → T_p = 0.256 µs → d = **51.2 m**.

Next: Chapter 7 — Ethernet (frame format, hubs, switches, collision domains).